

**Agentic Artificial Intelligence
(AI4015)**

Course Instructor(s):

Dr. Usama Imtiaz

Section(s): BSAI(A), BSDS(A,B)

Sessional-II Exam

Total Time (Hrs): 1

Total Marks: 60

Total Questions: 6

Date: Apr 16, 2026

Roll No

Course Section

Student Signature

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Attempt all the questions.

CLO 1: Analyze the fundamental concepts of agentic intelligence, including agent architectures, reasoning frameworks, and agent environment interactions

Q1: MCQS

(23 marks)

1. In a single-agent multi-role system (Researcher → Analyst → Writer), which failure is MOST likely when no validation layer exists?

- A. Deadlock due to synchronous messaging
- B. Role interference due to shared context
- C. Message duplication across agents
- D. Retrieval collapse

2. Which coordination strategy BEST fits a system where agents bid to execute tasks based on capability?

- A. Blackboard model
- B. Hierarchical coordination
- C. Contract Net protocol
- D. Sequential pipeline

3. In a decentralized MAS (swarm), debugging becomes difficult primarily because:

- A. Agents share memory
- B. No central control exists
- C. Messages are synchronous
- D. Embeddings are inconsistent

4. Which pattern is MOST appropriate when multiple agents process independent subtasks simultaneously and results are merged?

- A. Router
- B. Sequential
- C. Parallel + Aggregator
- D. Loop

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5. Which scenario BEST demonstrates role interference?

- A. Two agents communicate asynchronously
- B. Analyst validates Researcher output before writing
- C. Writer uses unverified Researcher data due to shared memory
- D. Supervisor enforces ordering

6. Which field ensures conversation continuity across multiple agent messages?

- A. sender
- B. payload
- C. conv_id
- D. performative

7. A system uses HTTP endpoints for agent communication. This implies:

- A. Tight coupling
- B. Physical isolation
- C. Shared memory
- D. Centralized orchestration

8. Which problem is directly solved by MCP but NOT by A2A?

- A. Message routing
- B. Agent identity
- C. Context size reduction
- D. Task decomposition

9. Which issue arises when agents send full data instead of MCP pointers?

- A. Deadlock
- B. Message overflow and inefficiency
- C. Loss of autonomy
- D. Reduced concurrency

10. Why is attention NOT considered long-term memory?

- A. It uses embeddings
- B. It is stored in vector DB
- C. It exists only within the context window
- D. It uses tokens

11. Which memory type stores “failed refinements and past decisions”?

- A. Working memory
- B. Semantic memory
- C. Episodic memory
- D. Buffer memory

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12. Context poisoning occurs because:

- A. Retrieval is slow
- B. LLM trusts retrieved data blindly
- C. Embeddings are inaccurate
- D. Tokens are limited

13. Retrieval collapse happens when:

- A. Too many documents are retrieved
- B. Same document dominates retrieval results
- C. Embeddings are missing
- D. Context window overflows

14. Semantic dilution is MOST likely when:

- A. top_k is too low
- B. top_k is too high
- C. embeddings are normalized
- D. chunk size is small

15. In RAG, retrieval happens BEFORE generation to:

- A. reduce token size
- B. improve reasoning
- C. ground the model with external knowledge
- D. speed up inference

16 Which index is BEST for hierarchical summarization of long documents?

- A. Vector Index
- B. Tree Index
- C. Keyword Index
- D. List Index

17. Which failure occurs when relevant information is split across chunks?

- A. Context poisoning
- B. Chunk boundary problem
- C. Retrieval collapse
- D. Embedding drift

18. Consider:

results = retriever.invoke(query)

What does this represent in agentic architecture?

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- A. Planning
- B. Memory recall (semantic memory)
- C. Execution
- D. Communication

19. A RabbitMQ broker in MAS primarily handles:

- A. embedding computation
- B. message routing
- C. tokenization
- D. context summarization

20. If an agent receives:

`{"type": "request", "payload": "Find 3 papers"}`

What is the expected behavior?

- A. Store information
- B. Execute task
- C. Reject message
- D. Broadcast message

21. If an agent repeatedly refines using same retrieved context, the likely issue is:

- A. Context poisoning
- B. Retrieval collapse
- C. Deadlock
- D. Overfitting

22. If top_k=1 in a RAG system, what is the MOST likely issue?

- A. Semantic dilution
- B. Lack of diversity and missing context
- C. Token overflow
- D. Embedding drift

23. If chunk size is reduced, what must typically be adjusted?

- A. embedding dimension
- B. similarity_top_k
- C. learning rate
- D. token limit

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CLO 2: Apply programming techniques to develop intelligent agents that incorporate reasoning, memory, and tool integration

Q2: Message Flow Analysis

(2+2+2 Marks)

```
{ "sender": "Planner", "receiver": "Researcher", "type": "request", "payload": "Find 3 papers", "conv_id": "c1" }
{ "sender": "Researcher", "receiver": "Planner", "type": "inform", "payload": "3 papers found", "conv_id": "c1" }
{ "sender": "Planner", "receiver": "Writer", "type": "request", "payload": "Write report", "conv_id": "c1" }
```

Questions:

- Identify the coordination flaw in this workflow.
- Which MAS principle is violated?
- Suggest one architectural fix using agents.

CLO 3: Design multi-agent workflows using modern frameworks and platforms to demonstrate coordination and RAG-based capabilities.

Q3 RAG Failure Diagnosis

(2+2+2 Marks)

A system uses:

```
retriever = index.as_retriever(similarity_top_k=10)
```

Observed behavior:

- Output becomes generic
- Irrelevant context dominates

Questions:

- Identify the failure type.
- Explain the root cause.
- Provide one mitigation strategy.

CLO 3: Design multi-agent workflows using modern frameworks and platforms to demonstrate coordination and RAG-based capabilities.

Q4 Memory Architecture Reasoning

(2+2+2 Marks)

A system uses:

- Context window only
- No buffers

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- No vector DB

Questions:

- a) Which memory layers are missing?
- b) What limitations will occur in multi-step tasks?
- c) Which component enables cross-session recall?

CLO 3: Design multi-agent workflows using modern frameworks and platforms to demonstrate coordination and RAG-based capabilities.

Q5 MAS Coordination Trace

(2+2+2 Marks)

A system has:

- Planner
- Researcher
- Critic
- Writer

Observed issue:

Writer produces output before Critic validation.

Questions:

- a) What coordination failure occurred?
- b) Which pattern should be enforced?
- c) Which framework feature can fix this?

CLO 4: Evaluate the effectiveness and reliability of agentic systems through debugging, observability methods, and performance metrics.

Q6 Faulty Agentic Research Pipeline

(4+2+2+5 Marks)

A student builds the following agentic system for report generation.

```
shared_memory = {}
def planner(query):
    shared_memory["goal"] = query
    return retriever(query)
def retriever(query):
    docs = vector_db.search(query, top_k=8)
    shared_memory["docs"] = docs
    return analyst(docs)
```

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```
def analyst(docs):  
    summary = llm("Summarize these documents: " + str(docs))  
    shared_memory["analysis"] = summary  
    return writer(summary)  
  
def writer(summary):  
    report = llm("Write final report using: " + summary)  
    return report
```

Observed Execution Trace

```
[Planner] Goal received: "Write a report on AI adoption in healthcare"  
[Retriever] Retrieved 8 chunks  
[Retriever] Top results include:  
    Doc 1: "AI in healthcare, 2021 trends"  
    Doc 2: "Hospital automation overview"  
    Doc 3: "AI ethics in education"  
    Doc 4: "AI in healthcare, 2021 trends" ← repeated  
    Doc 5: "Medical imaging with AI"  
    Doc 6: "AI in healthcare, 2021 trends" ← repeated  
[Analyst] Summarizing retrieved documents...  
[Writer] Generating report...  
[Output] Final report includes outdated 2021 statistics and one paragraph about education ethics
```

Additional System Change

The student now adds another agent:

```
def critic():  
    if "education" in shared_memory["analysis"]:  
        return "Irrelevant domain detected"  
    return "Looks good"
```

But the updated pipeline is:

```
def analyst(docs):  
    summary = llm("Summarize these documents: " + str(docs))  
    shared_memory["analysis"] = summary
```

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```
return writer(summary)

def writer(summary):
    report = llm("Write final report using: " + summary)
    feedback = critic()
    return report
```

Questions

a) Why did the final report include:

- outdated statistics
- irrelevant education content

Your answer must connect retrieval behavior to downstream reasoning.

- b) Which major coordination/design flaw exists in the updated pipeline with the critic() agent?
- c) Why is the critic ineffective even though it exists in the system?
- d) Rewrite the pipeline logic in correct high-level pseudocode order so that validation happens before final report generation.

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